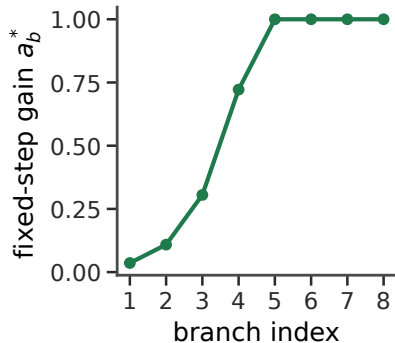
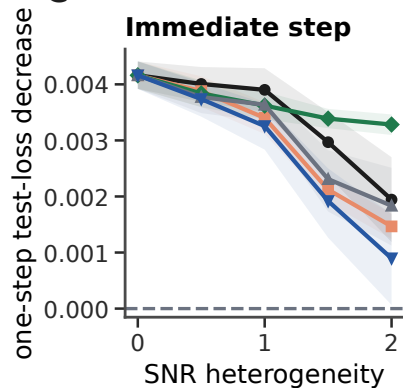


A**Physical isolation**

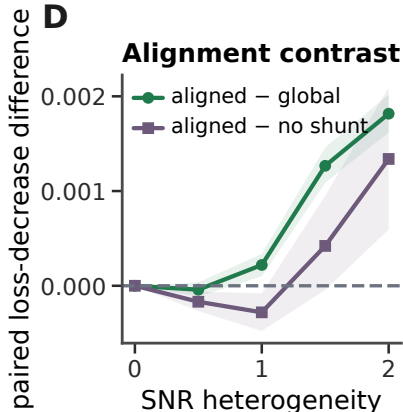
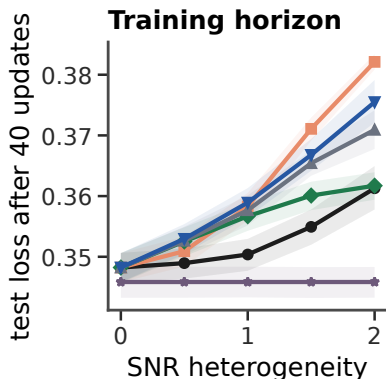
shunt $\kappa_b \geq 0$ + clamp current $\kappa_b V$

$V' = V$ but eligibility $\times G/(G + \kappa_b)$

isolates physical input-resistance attenuation

B**Step-consistent profile****C****Immediate step**

—●— no shunt —■— best global —◆— SNR-aligned —▲— shuffled —▼— anti-aligned —◆— exact clean BP

D**Alignment contrast****E****Training horizon****F****Final boundary**